Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 12.1 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 1 Class 12.1 AVSummary

1 UE26ME142A Unit 1 Class 12.1 AVSummary

Study focus: UE26ME142A_Unit 1_Class 12.1_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000270 · EV0000271

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Venkitish from Civil Engineering under Department of Science and Humanities, PES University.
- This is class number sixteen.
- In the last class, that is class number fifteen, we had discussed.
- Two problems taken from the unsolved problems from the prescribed textbook that is Engineering Mechanics Statics by Miriam, Craig and Bolton.
- And it is the eighth edition.
- Today we will be continuing with.
- Some more numericals.
- And this is the last topic from unit one.
- For today's discussion, I have considered a problem on 2 bar 98.
- The statement of the problem goes something like this.
- Five forces are applied to the beam, trolley as shown, determine the coordinates of the point on the y-axis through which the standalone resultant R must pass.
- If F is equal to 5 kilonewton and theta is equal to 30 degrees.
- The Y intercept of the standalone resultant.
- What are we supposed to find is the y-intercept or the coordinate of the point on the y-axis at which the standalone resultant needs to be applied.
- This force the magnitude is three kilonewton again, it is acting along the x axis to the left.
- Then coming to this force, the magnitude of the force is 0.6, again it is acting along y, but it is in towards the right, so positive x axis, then this force is 0.8 kilometers, again it is acting

along the x axis, again it is positive because it is acting to the right.
- The angle at which the F force is acting with respect to the y-axis.
- Axis, you are supposed to take that as the positive axis.
- X axis positive is to the right and downwards is the positive y axis.
- Of course, your x-axis is the center of this, which is at 140 mm.
- how do we go about this?
- This is a free body diagram.
- Then the F force is acting here.
- Then of course this also is considered, which you are unable to see here.
- This is the free body diagram here.
- This distance is 280 mm.
- First take all the forces along acting along x axis, algebraic sum of all the forces.

Example or application from the lecture

- The force F magnitude is given as 5 kilo Newton and the angle T at which this force is acting is given as 30 degrees.
- F force, the magnitude of F is given as 5 kilometer.
- Is given as 30 degrees.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000270 · EV0000271 · EV0000272 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then take all the forces acting along y, algebraic sum of that, then you get the component of the resultant about the y-axis.

Key details

- With the values that we have here, you get the component of the resultant force acting towards the left and along the x-axis and component of the force.
- Along y-axis will be acting downwards.
- You can also find out the angle at which the resultant is acting with respect to the negative x axis or the positive y axis.
- Once you get that.
- The next thing is we have to find out where are we supposed to apply this force along the y-axis for which we will go for usage of principle of movements or extension of Baring North principle, that is.
- Moment produced by all the forces about one point, where in this problem we will take it as a point O should be equal to the moment produced by the resultant force.
- If I take the resultant to be acting here, I will get the X component of the resultant into the distance at which it is acting.
- If I take it as y, then I can find out y.
- Then we have one force acting towards the right, that is point six, another force acting towards the right again, that is point eight.
- Then we also have the component of the.

- F force that is acting along X axis, which is B 5, sine 30 degrees, that will be acting to the right, so that is positive.
- Next, to take all the forces acting along y, f p will be.
- The component of the F4 which is acting along the y axis which will be downward that is positive, which will be phi cos 30 degrees, you will get that as 4.33 kilonutons.
- Now take the moment about the point O because of all the forces.
- You take the force 4 kilometer that is acting to the left side along the x-axis, the distance would be 280 divided by 2, which was the distance which I had talked about, plus we should take the radius of the pulley that we have or the diameter of the pulley at the top as well as at the bottom.
- Then plus 3 kilonutan is acting at the bottom of the pulley, top pulley.
- Plus we have 0.6, which is also going to be acting at the downward, which is at a distance of 280 by 2 itself.
- This would also produce an anti-clockwise movement.
- Plus 0.8 would also produce an anti-clock point moment, but the distance would be 280 by 2 plus 55 plus 55.
- Because I have taken this in the millimeters on this force magnitude in kilone, so it is kilonewton millimeters.
- Applied this resultant force.
- On the x-axis, okay.
- To the left of O, then you will have a negative movement, so you get that to be equal to 393.53 mm.
- Or I'll apply this resultant force, this resultant force on the y-axis, which is below O.

Example or application from the lecture

- figure that is given you can make out what is the radius of the pulley.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000274 · EV0000275 · EV0000276 · +1 source units in pack.json

Core explanation · 10:00–14:59

Minus this would be minus 3.1, which will be acting to the left.

Key details

- That would be the.
- Y intercept of the resultant from the point O.
- Next, I have taken the last problem in this unit, unit 1, chapter 2.
- That is problem 2 bar 100.
- The problem statement goes something like this.
- An exhaust system for a pickup truck is shown in the figure.
- The weights WH, WM under W1 of the head pipe.
- Moffl and tailpipe are 10, 100, comma 50 respectively.
- And at the indicated points, if the exhaust pipe hanger.
- At point A is adjusted so that its tension FA is 50 newtons.

- Determine the required forces in the hangar at point B, C and D.
- Why is a zero force couple system at O is desirable.
- The weight of WH, which is this weight, the weight of W M, which is this one, the weight of W1, which is this one.
- And they have asked you to determine the force at B, that is this force F B, F C, and F such that the force couple system at this point O is equal to 0.
- If the exhaust pipe hanger at point A is adjusted so that its tension FA is 50 newton, why is the zero force couple system what most desirable?
- That is what is asked.
- This would be a candile.
- If the forces are balanced with each other, there won't be any forces that is acting at war.
- That is the first thing why a force couple system, zero force couple is desirable.
- And there is no force acting at this point.
- And it should also not rotate.
- That is why your couple at this point should also be zero.
- That is the explanation for why we should not.
- Have or have or why we should have zero force couple system.
- And also we can note down that the force acting at AB, then your WH, WM, and WT, all of them are acting vertically.
- Except that F C and F D are acting.
- At an angle to the vertical, which is 30 degrees for both of them.
- That means the component of the resultant force along X and along Y should be zero.

Example or application from the lecture

- Or it is W T it should be, I think I'm sorry, this is W T, so this should be W T are given respectively as 10, 100 and 50.
- They have also given the force acting at A, that is this force F A to be equal to 50 newton.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000278 · EV0000279 · EV0000280 · +1 source units in pack.json

Core explanation · 15:00–19:59

X axis force, whereas Fd will produce a positive x axis force.

Key details

- Next, if I take all the forces acting along the vertical.
- I can say that this component F C and F D plus this component F C minus these three forces would give me the force in F B.
- Then to determine the value of F C and F D, I will take the moment about O, which we will get it in terms of F P, F C and F D.
- One is force at B, which we have to find, force C we have to find, force at B we have to find.
- There are three unknowns, so at least we require three equations.
- Which are these three equations?
- Since the resultant at O is 0, the component of the resultant along X should be equal to 0.

- The rest is acting along the vertical.
- Minus 10 is your one force that is W acting.
- Then one more force.
- W this is WH this is W M and this is W T all the three of them are acting downwards so it is negative Your F B is acting upwards.
- The component of the force F C along the vertical will be acting upwards, so positive, that is F C cos 30 degrees.
- The component of the force F T will also be acting upwards, that is F T cos 30 degrees acting upwards.
- From this, if I take all the components acting along the y, I can easily make out that F C is equal to F D, which is equal to F.
- If I take all the component acting along the y direction, substituting this here for F C and F D as F, I can get the value of F P in terms of F.
- To all the unknowns to be acting or in terms of F, I can make out.
- Next you take movement about O of all the forces.
- 50 is the force acting at A, which is acting upwards.
- 100 is M B which is acting downward, which will produce a clockwise movement, the distance from O to B or where the force is acting F.
- f you will have the distance as 1.35 then your f p force is acting upward which will produce an anti clockwise movement the distance from o to that force is two kilon two meters so that is what is taken here then your f.

Example or application from the lecture

- As per the statement given, we are supposed to find out the force at B, force at C, and force at D.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000282 · EV0000283 · EV0000284 · +1 source units in pack.json

Core explanation · 20:00–24:59

T is acting downward, which will produce a clockwise movement.

Key details

- Then we have the two forces F C and F D, the component of those two, you can take it as 2F, cos 30 degrees acting vertically, the distance of that would be 2.9.
- Here I can substitute this in this equation, and then I can get the value of F B.
- Then substitute that whatever relationship that we have got, F B in terms of F, then substitute.
- Here we get the value of F C and F D.
- I hope this is also clear.
- In the next class, we will have an experiential component class, that is class number 17.
- Next to that, we will start with the next unit, that is unit 2.
- That unit is equilibrium chapter.
- I am Dr.

- SV Ventesh from Civil Engineering under Department of Science and Humanities, PS University.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000286 · EV0000287 · EV0000288 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.
4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.
9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.

13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00062	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00063	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00064	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00065	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

C Revision and retrieval practice